

CS 162

Computer Architecture

Final Review

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Material on exam



- **Slides 8 to 13**
- Understand memory hierarchy
- Understand thread parallelism
- Understand data parallelism

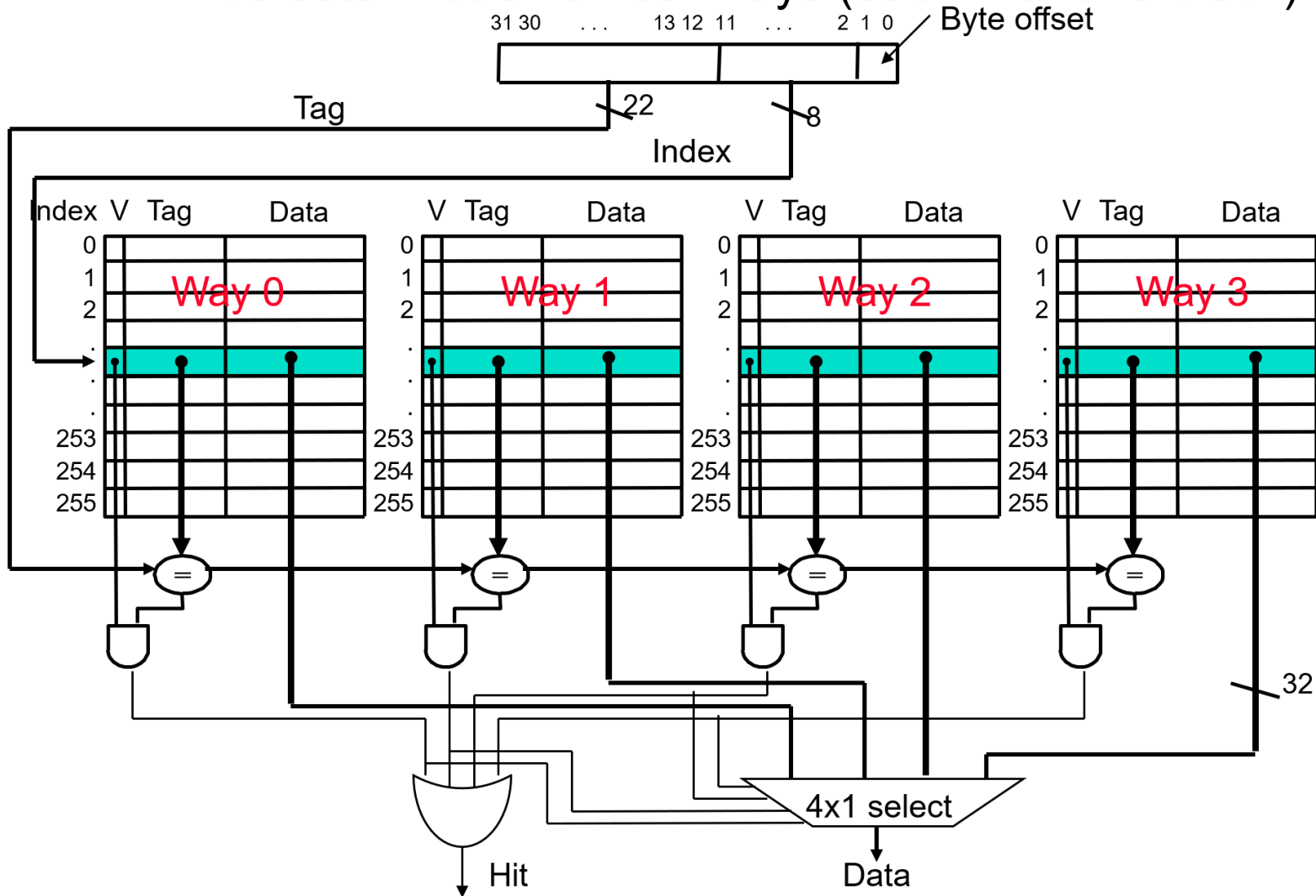
Material on exam



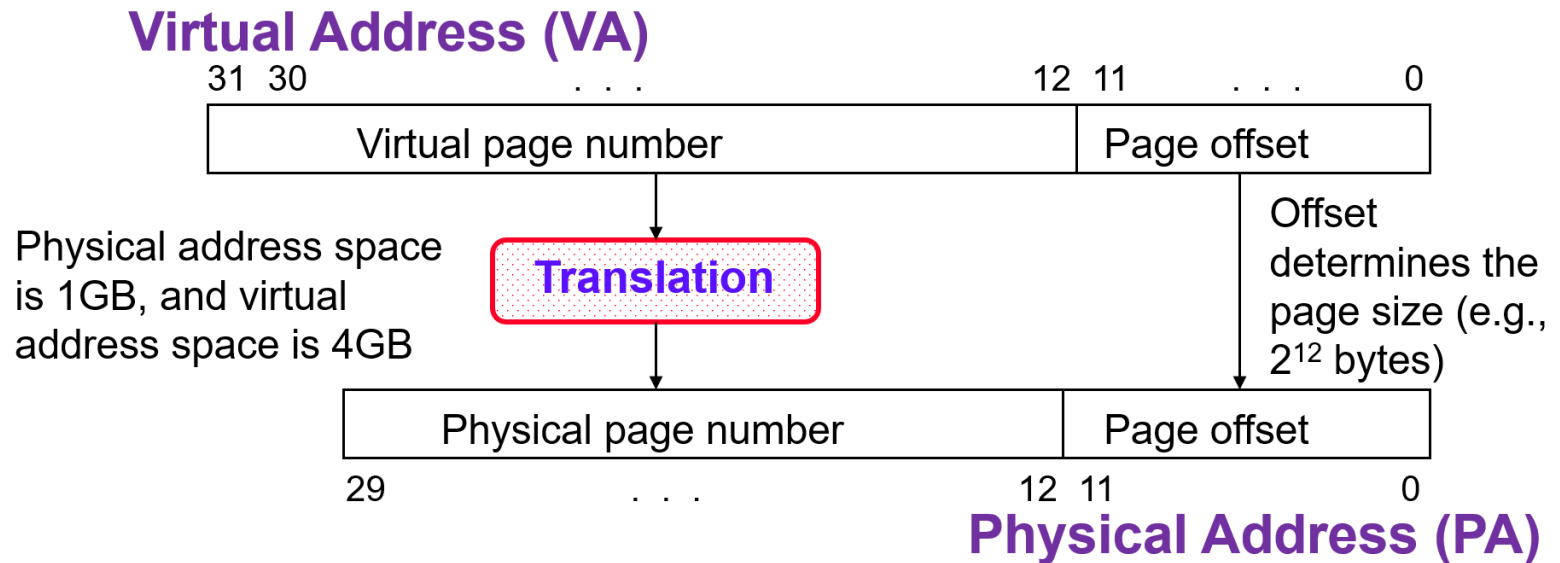
- Understand memory hierarchy

Set Associative – 4KB Cache

□ $2^8 = 256$ sets in each of four ways (each with one block)



Address Translation



Classification of cache misses



- **Compulsory Misses**: Sad facts of life. Example: cold start misses.
- **Capacity Misses**: Increase cache size
- **Conflict Misses**: Increase cache size and/or associativity.

Basic ways to reduce miss rate (§ B.3)



- Which of the three C's are affected by the the following cache optimizations?

1. Larger Block Size

2. Larger caches

3. Higher Associativity

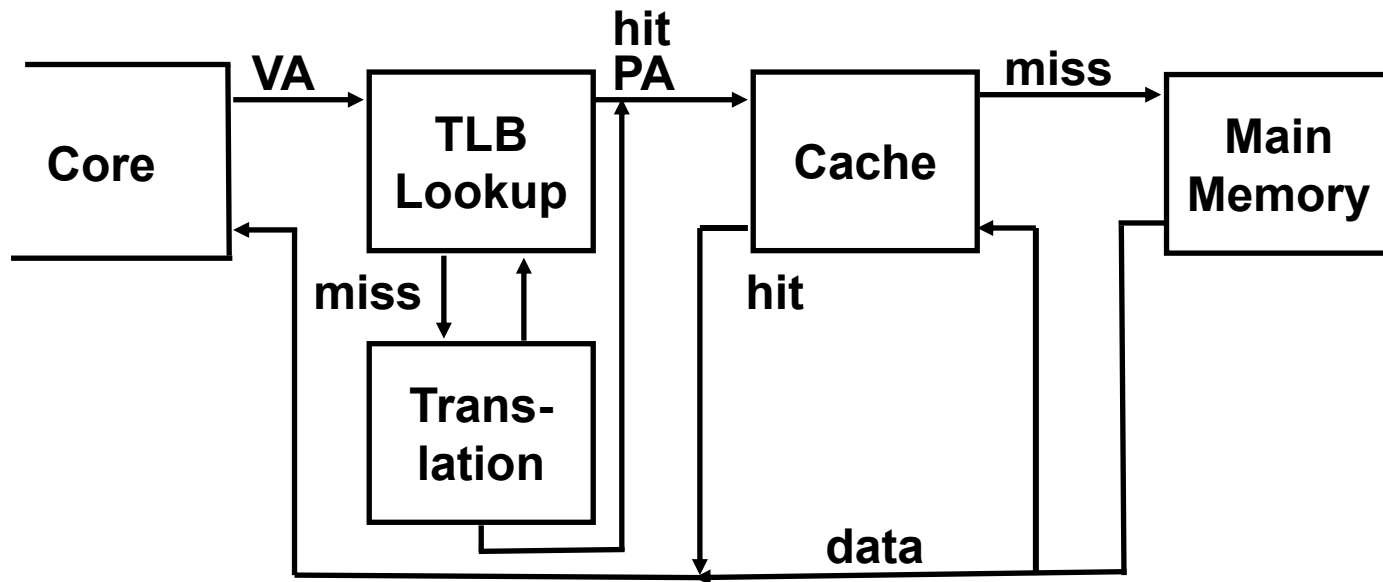
Compute average memory access time

- Miss rate of L1 = 4%, acc time is one cycle
- If L2 is set-associative, local miss rate = 20% and acc time = 11 cycles
- Miss penalty for L2 is 100 cycle.

$AMAT = L1 \text{ hit time} + L1 \text{ miss rate} \times (L2 \text{ hit time} + L2 \text{ miss rate} \times L2 \text{ miss penalty}).$

$$AMAT = 1 + 0.04 (11 + 0.20 * 100) = 2.24 \text{ cycles}$$

A TLB in the Memory Hierarchy



Example (No overlap of TLB and L1 \$)

Unit	Access Latency	Hit Rate
TLB	1 cycle	95%
L1	1 cycle	95%
L2	8 cycles	80%
L3	50 cycles	50%
Memory	100 cycles	100%
Page table access & TLB update	200 cycles	100%

Cache Optimizations

- 1. Small and Simple Caches to reduce hit time**
- 2. Way prediction to reduce hit time**
- 3. Pipeline cache access to increase bandwidth**
- 4. Non-blocking caches to increase bandwidth**
- 5. Multi-bank caches to increase bandwidth**
- 6. Critical word first + early restart to reduce miss penalty**

Cache Optimizations

- 7. Merging write buffer to reduce miss penalty**
- 8. Compiler optimizations to reduce miss rate**
- 9. Hardware prefetching to reduce miss rate and penalty**
- 10. Compiler prefetching to reduce miss rate and penalty**
- 11. Using victim caches**
- 12. Computing with Near Data**

TLB Optimizations

- 1. Large pages**
- 2. TLB compression**
- 3. Speculative TLB**
- 4. Range TLB/ Coalescing**
- 5. Eager paging**

Material on exam



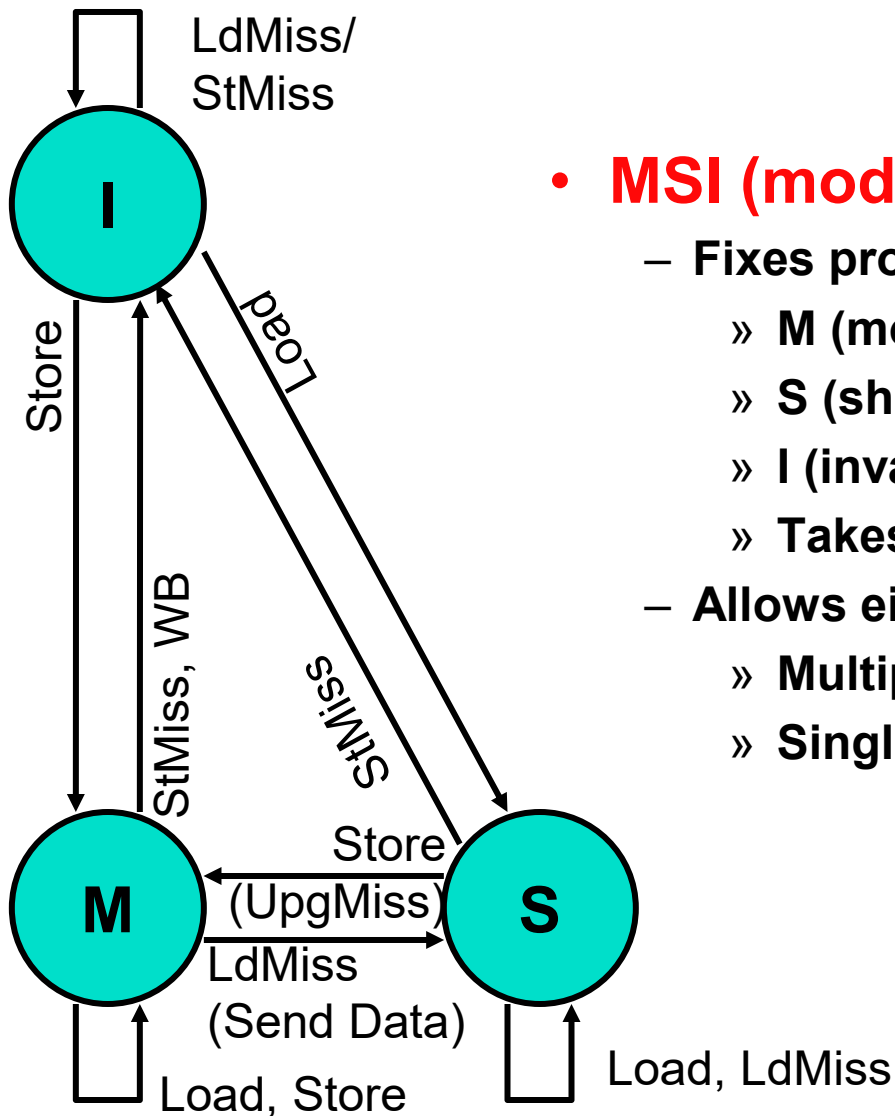
- Understand thread parallelism

A Coherent Memory System



- Any read of a data item should return the most recently written value of the data item
 - Coherence – defines **what values** can be returned by a read
 - Stores to the same location are **serialized** (two stores to the same location must be seen in the same order by all cores)
 - Consistency – determines **when** a written value will be returned by a read

MSI



- **MSI (modified-shared-invalid)**

- Fixes problem: splits “V” state into two states
 - » M (modified): local dirty copy
 - » S (shared): local clean copy
 - » I (invalid)
 - » Takes 2 bits to implement
- Allows either
 - » Multiple read-only copies (S-state) **--OR--**
 - » Single read/write copy (M-state)

MSI protocol state transition table

State	<i>From This Core</i>		<i>From Other Cores</i>	
	Load	Store	Load Miss	Store Miss
Invalid (I)	Miss → S	Miss → M	---	---
Shared (S)	Hit	Upg Miss → M	---	→ I
Modified (M)	Hit	Hit	Send Data → S	Send Data → I

- ❑ M → S transition on a LdMiss also updates memory (Sends Data) since the block is Dirty

Material on exam



- Understand data parallelism

SIMD Processing



- **Single instruction operates on multiple data elements**
 - In time or in space
- **Multiple processing elements**
- **Time-space duality**
 - **Array processor**: Instruction operates on multiple data elements at the same time
 - **Vector processor**: Instruction operates on multiple data elements in consecutive time steps

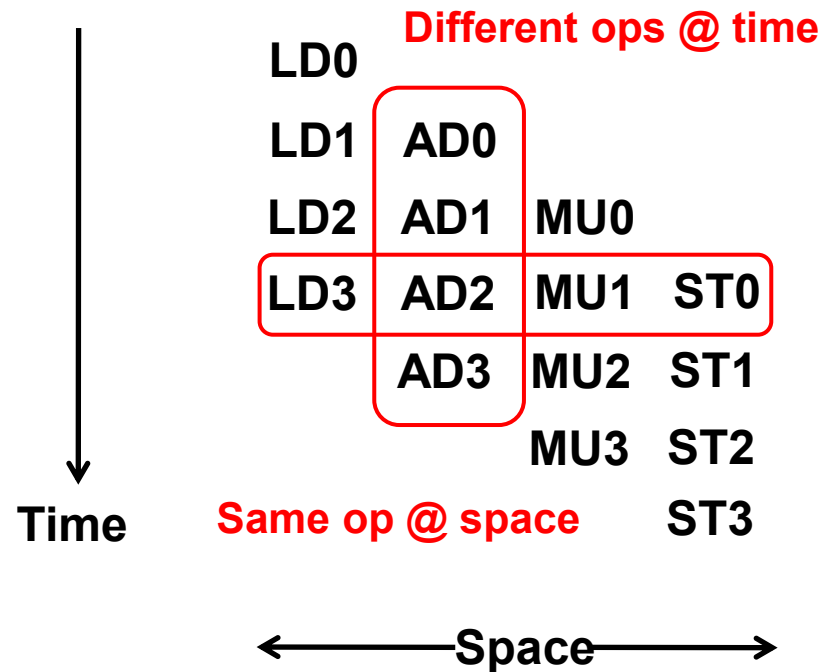
Vector Processors

VECTOR PROCESSOR



Instruction Stream

LD VR $\leftarrow$ A[3:0]
ADD VR $\leftarrow$ VR, 1
MUL VR $\leftarrow$ VR, 2
ST A[3:0] $\leftarrow$ VR



Parallelizable / Vectorizable



- **Example:**

```
for (i=2 ; i<1000 ; i++)  
    x[i] = x[i-1] + x[i-2];
```

- **Example:**

```
for (i=2 ; i<1000 ; i++)  
    x[i] = x[i+1] + x[i+2];
```

Final Exam

Location: **WCH143**

Time: **Mar. 6 and Mar. 9 class time**

Duration: **55 mins**

No textbook, no cheat sheets, no calculator

- **Please submit End-of-term evaluations >80% 3 bonus points**